SYLLABUS ECEN 635 Fall 2025

Course: ECEN 635 Advanced Electromagnetic Fields
Prerequisites: ECEN 322, Properties of Plane Waves, BasicTransmission Lines
Instructor: Robert Nevels, Rm. 205K Wisenbaker (WEB)
Text: *Advanced Engineering Electromagnetics*, 3rd Ed. C. A. Balanis, *Handbook of Mathematical Functions* by M. Abramowitz and I.A. Stegun. There is an on-line version of this handbook on the web and an inexpensive paperback edition.
Class Time: 4:10 -5:25 PM MW
Location: Room: 49 WEB
Credit Hours: 3
Office Hours: Thursday 3:30-5:00 PM in WEB
Zoom Office Hours: Thursday 8:00-9:00
https://tamu.zoom.us/j/93373434673?pwd=LrHjPXKAbVyM9y4hyE9ON3FPHXJKKs.1
Meeting ID: 933 7343 4673
Passcode: 288526

Section 1

Maxwell's equations time and frequency differential and integral	1.1-1.2, 1.7
Continuity condition	1.5.1-1.5.3,1.7.2
Constitutive relations	1.3
Material parameters	2.2, 2.4.2,2.4.3,2.8
Helmholtz equation	3.3, 4.2.1
Plane waves	4.2.2,4.3
Poynting theorem	1.6, 1.73
Polarization	4.4

Section 2

Boundary Conditions	1.5
Waves in dielectrics and conductors	5.2
Wave interaction at boundaries - critical angle, Brewster angle, duality	5.3
Transmission line models	Notes
Potentials and radiation	6.1,6.2,6.3,6.4
Radiation condition	6.6,6.7
Duality of electric and magnetic dipoles	6.8.1,6.8.2

Section 3

Reciprocity	7.5
Uniqueness theorem	7.3
Equivalence principle	7.8, 7.9
Image methods	7.4
Green's functions	Notes

Section 4

Rectangular waveguide with potentials	8.2, 8.3
Alternative mode sets	
Dielectric waveguides	8.7.1,8.7.2
Transverse resonance method	8.6,8.61,8.71, 8.7.2
Rectangular cavity	8.3, 8.3.1, 8.3.2
Rectangular microstrip antenna	Notes

Section 6

Cylindrical wave functions	9.2.1
Cylindrical waveguide	9.2.1, 9.2.3
Scattering from a wedge	11.6.1
Circular microstrip antenna	Notes

Section 7

Spherical wave functions	11.7.1, 11.7.2,
Scattering from a conducting sphere (Mei series)	11.8.1
Scattering from a dielectric sphere	11.8.2

Grading System:

	Weight:	
Midterm	40%	October 15th (Wednesday)
Final	40%	December 12 (Friday) 3:30AM-5:30PM
Homework	20%	
TOTAL	100%	

If your over-all percentile score falls into one of the following ranges, you are guaranteed at least the following indicated grade:

85% - 100%	A
70% - 84%	B
60% - 69%	C
50% - 59%	D

Homework: Homework, uploaded to the class ECEN 635 Canvas website, is due on Friday 11:59 PM unless otherwise announced. After class late homework will be accepted until Saturday 11:59 PM with a 10% penalty and not accepted after that except with a university approved excuse.

Makeup Exams: If you have a university approved excuse for missing an exam, your Final Exam score will be doubled to make up for the missed exam.